26900354 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEMATA01 - ADVANCED DATA ANALYSIS USING PYTHON

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer any **10** questions

Each question carries **3** marks

1. Tukey's HSD is generally performed: [K] / [CO1]

   a). Before ANOVA b). Instead of ANOVA in every situation

   c). Only when there are two groups d). After a significant ANOVA result

2. If a query returns 10 rows and fetchone() is called once, how many rows are retrieved by that call? [A] / [CO2]

   a). 0 b). 9 c). 1 d). 10

3. If p = 0, which component is absent? [A] / [CO3]

   a). AR b). MA c). Differencing d). Seasonal

4. Which function is used to calculate the hierarchical clustering linkage for the dendrogram in the example? [U] / [CO4]

   a). ward() b). cluster()

c). fit_predict() d). accuracy_score()

5. Consider stats.ttest_1samp(data, 1000). What does 1000 represent? [U] / [CO1]

a). Hypothesized population mean b). Sample size

c). Sample variance d). p-value

6. What does ndimage.maximum_filter(img,3) do? [U] / [CO1]

a). Uses the maximum value in a neighborhood b). Uses the minimum value

c). Uses the median d). Rotates the image

7. Which function displays the column names of a DataFrame? [K] / [CO2]

a). shape() b). fields() c). colum() d). keys()

8. Which technique primarily assigns exponentially decreasing weights to historical observations? [U] / [CO3]

a). EWMA b). Differencing

c). Seasonal decomposition d). Rolling standard deviation

9. Which type of Factor Analysis assumes that each factor is associated with a particular set of observed variables? [K] / [CO4]

a). EFA b). CFA c). PCA d). Cluster Analysis

10. A time series with a nonconstant mean over time is said to have a: [U] / [CO3]

a). Seasonal component b). Random component

c). Trend d). Moving average

11. The Mann–Whitney U test is generally used to compare: [K] / [CO1]

a). More than three groups b). Two independent groups

c). Two paired groups only d). Variances only

12. Which function assigns ranks to rows based on an ordering condition? [K] / [CO2]

a). COUNT() b). RANK() c). GROUP() d). ORDER()

13. What is the primary purpose of forecasting in business? [K] / [CO3]

a). To eliminate uncertainty

b). To support managerial decision-making

c). To increase historical data

d). To avoid planning

14. If a cluster contains 20, 25, and 30, its centroid is: [An] / [CO4]

a). 20 b). 30 c). 25 d). 8

15. A company wants to divide its customers into groups based on purchasing patterns, without having predefined customer categories. Which technique is most suitable? [A] / [CO4]

a). Linear regression

b). Principal Component Analysis

c). K-means clustering

d). Decision tree

[3x10 = 30]

Part B

One or Two Sentences

Answer any **4** questions

Each question carries **5** marks

16. Describe how ranks are assigned in the Wilcoxon Signed-Rank test when tied absolute differences occur. [U] / [CO1]

17. Explain the concept of a subquery in SQL. Describe the parent statement and subquery with a suitable example. [U] / [CO2]

18. What is RMSE? Explain its importance in model evaluation. [U] / [CO3]

19. Explain the concept of Principal Component Analysis (PCA). Discuss its main objective and how PCA helps in dimensionality reduction. [U] / [CO4]

20. Explain the Q-Q plot/probability plot and its interpretation. [U] / [CO1]

21. Explain the role of rolling mean and rolling standard deviation in determining the stationarity of a time series. [U] / [CO3]

[5x4 = 20]

Part C

Short Essay Questions

Answer any **2** questions

Each question carries **10** marks

22. Explain special subpackage and the purpose and usage of cbrt(), comb(), exp10(), exprel(), gamma(), lambertw(), logsumexp(), and perm() with suitable Python examples. [A] / [CO1]

23. Given a table named cars1, write SQL queries to create a table named cars1 in a datbase and to calculate the total number of cars, total MPG, average MPG, highest MPG, lowest MPG, and number of distinct cylinder types. Explain each query and the expected type of result. [A] / [CO2]

24. Explain the methods used to make a time series stationary, using seasonal decomposition, and differencing. Give suitable examples and Python code. [U] / [CO3]

25. Using Python, perform Factor Analysis on the 10 independent variables of the Diabetes dataset using three factors and Varimax rotation. Write a program to create and fit the Factor Analyzer model, display the factor loadings, and calculate the variance explained by each factor. Explain the purpose of n_factors, rotation, fit(), loadings_, and get_factor_variance(). Finally, explain how the factor-loading matrix can be used to group and interpret the observed variables. [An] / [CO4]

[10x2 = 20]